

Forge: A Cartesian Immersed-Boundary Approach to Unsteady Transonic Flow

Without Turbulence Modeling, Wall Functions, or Artificial Dissipation

Solver	Description
Algorithm	Semi-Lagrangian advection + spectral pressure projection
Turbulence	None (implicit LES)
Wall treatment	Physics-based molecular interface model
Precision	float64 throughout, PyTorch GPU
Hardware	NVIDIA RTX A4000 (16 GB), single GPU
Implementation	~3,200 lines Python (one file)

[OAT15A Flow
Visualization]

Globally-synchronized explicit timestep

1. Freeze entire field state at time t
2. Compute ALL domain tendencies simultaneously from that snapshot
(fluid, thermal, structural, electromagnetic — all see same state)
3. Sum tendencies and project forward once

No iterative coupling between domains

No matrix inversions or inner iterations

No sequential update contamination

**Unconditionally stable: $CFL > 10^{31}$
recorded without divergence**

Solver Components

Component	Approach
Advection	Semi-Lagrangian (3D trilinear backtrack)
Pressure	Spectral (FFT) Leray projection
Wall BC	Molecular accommodation interface
Turbulence	None — implicit LES from scheme
Dissipation	Zero artificial dissipation
Density	Active transport + equation of state
Diagnostics	Proprietary info-theoretic scalar

How does it work without a turbulence model?

The wall is not a mathematical boundary condition.
It is a physics-based interface between two materials.

Momentum exchange is governed by the molecular accommodation coefficient between the fluid and wall material. The velocity field near the wall develops naturally through that exchange — outward through advection and diffusion. No prescribed velocity profile.

Grid considerations:

Uniform Cartesian. No stretching, no clustering, no aspect ratio management. The consideration is cells per chord and domain extent. Surface interaction is in the interface physics, not in the mesh.

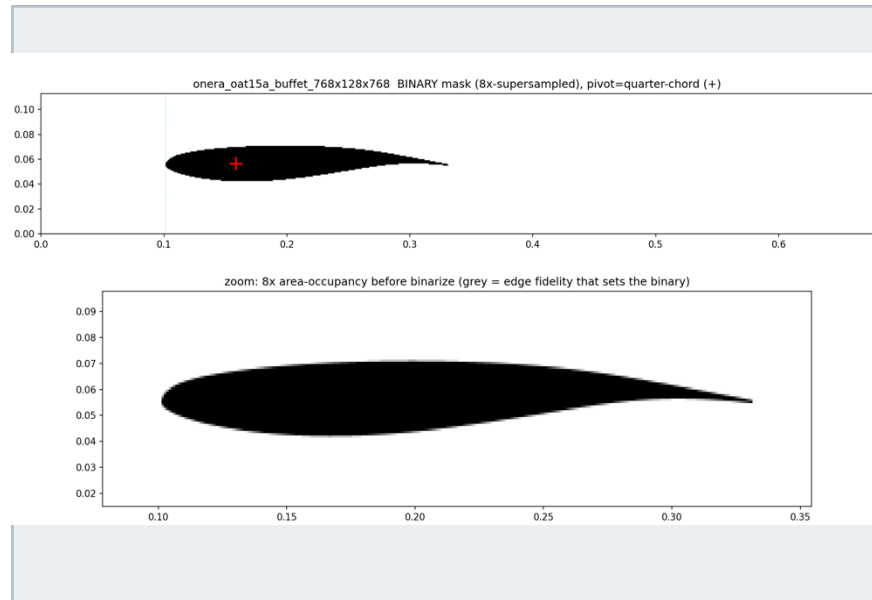
Conventional vs. Forge

	Conventional	Forge
Wall BC	Wall function or $y^+ < 1$	Molecular interface
BL mesh	52+ layers $s=1.2$ growth	Uniform Cartesian
Turb model	SA, SST $k-\omega$, DDES, etc.	None
Dissipation	Flux limiters, artificial visc.	Zero
Grid effort	Weeks	Minutes



ONERA OAT15A Grid Specification

Parameter	Value
Grid	768 × 512 × 16
Total cells	6,291,456
dx, dy, dz	1.198, 1.797, 1.438 mm
Domain	0.92 × 0.92 × 0.023 m
Chord (c_ref)	230 mm (model scale)
Cells/chord	~192 (streamwise)
Surface samples	17,136
Span cells	16 (10% chord)



ONERA OAT15A: Test Case Setup

Forge Physics LLC

Condition	Value	Source
Mach	0.7303	DPW-8 TC1
Angle of attack	3.78°	Geometry rotated in conversion°
U_∞	241.0 m/s	From Mach + T_{static}
ρ_∞	0.923 kg/m ³	Experimental condition
T_{static}	271.0 K	Experimental condition
q_∞	26,804 Pa	$\frac{1}{2}\rho U^2$
c_{sound}	330.0 m/s	$\sqrt{(\gamma RT)}$
dt	1.0×10^{-4} s	Explicit timestep
Steps analyzed	750 (steps 751–1500)	Post-transient window
Boundary	Tunnel mode, open walls	Inlet/outlet + open lateral

Note: OAT15A geometry per ONERA designation. Model-scale chord = 230 mm.

Quantity	Mean	Range (unsteady)
CL	0.1908	0.186 – 0.193
CD	0.0389	0.036 – 0.041
CM	0.3064	0.297 – 0.313

Diagnostic convergence (D = 1 - ISC_analog):

Step	D	Status
0	0.9997	Initial (uniform)
1	0.0632	95% resolved
5	0.0075	99.3% resolved
1500	0.0227	Settled limit-cycle

Spectral characteristics:

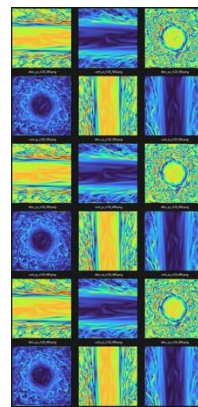
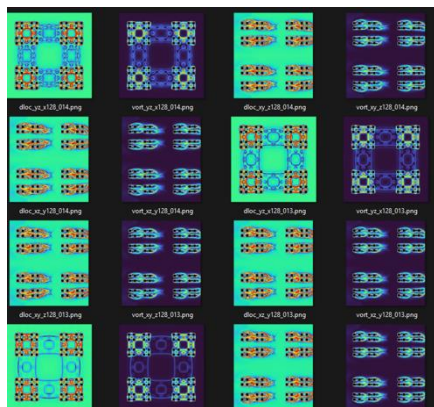
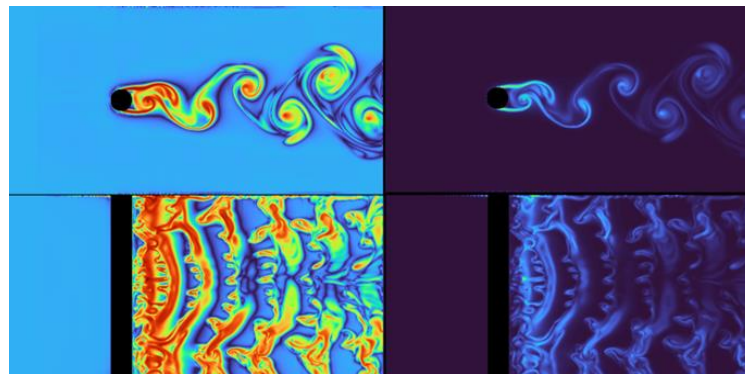
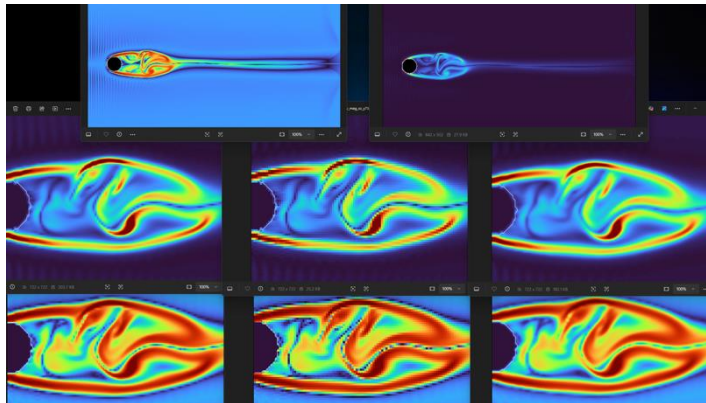
Probe location: $x/c = 0.45$
(upper surface, mid-chord)

Analysis window: 750 steps
 $dt = 1.0 \times 10^{-4}$ s
Nyquist: 5,000 Hz

Peak PSD at 13.3 Hz
(0.0112 Pa²/Hz)

Broadband content above 100 Hz
consistent with turbulent buffet

Reference: Jacquin et al. 2009



What we showed:

A Cartesian immersed-boundary solver that resolves unsteady transonic flow — including shock buffet — without turbulence modeling, wall functions, or artificial dissipation. Running on consumer workstation hardware.

Next steps:

Grid refinement studies on ONERA OAT15A
Full AoA sweep for buffet onset characterization
NASA CRM force/moment polar completion
Aeroacoustic analysis from $\text{div}(\mathbf{u})$ field data

Post-workshop:

Interested in aligning with the Source of Scatter Working Group and the Buffet Working Group for continued collaboration.

Thank you

Ben Eugster
Forge Physics LLC

Questions welcome.